PROJECT REPORT: CUSTOMER SEGMENTATION ANALYSIS

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1. INTRODUCTION & DATASET OVERVIEW

This project focuses on customer segmentation using data analytics to understand customer behavior and support targeted marketing strategies for improved customer retention and business growth.

The analysis is performed on the “Mall Customer Segmentation Dataset” sourced from Kaggle, consisting of 200 customers. The dataset includes key attributes such as Age, Annual Income (k$), Spending Score (1–100), and Sales Target.

The objective is to identify meaningful customer groups based on behavioral and demographic patterns.

2. METHODOLOGY

The analysis combines Power BI for data visualization and Python (Scikit-learn) for machine learning-based clustering.

K-Means clustering was applied after standardizing the dataset to ensure equal weightage across all variables. The optimal number of clusters was determined using the Within-Cluster Sum of Squares (WCSS) method.

$WCSS = \Sigma ||x - \mu||^2$

This approach helps minimize variance within clusters and improves segmentation accuracy.

3. POWER BI DASHBOARD INSIGHTS

An interactive Power BI dashboard was developed to explore and analyze customer data.

Key KPIs:

• Total Customers: 200

• Average Age: 38.85 years

• Average Annual Income: $60.56k

• Average Spending Score: 50.20

Visualizations Used:

• KPI cards for summary metrics

• Scatter plot for Income vs Spending Score

• Bar charts for demographic distribution

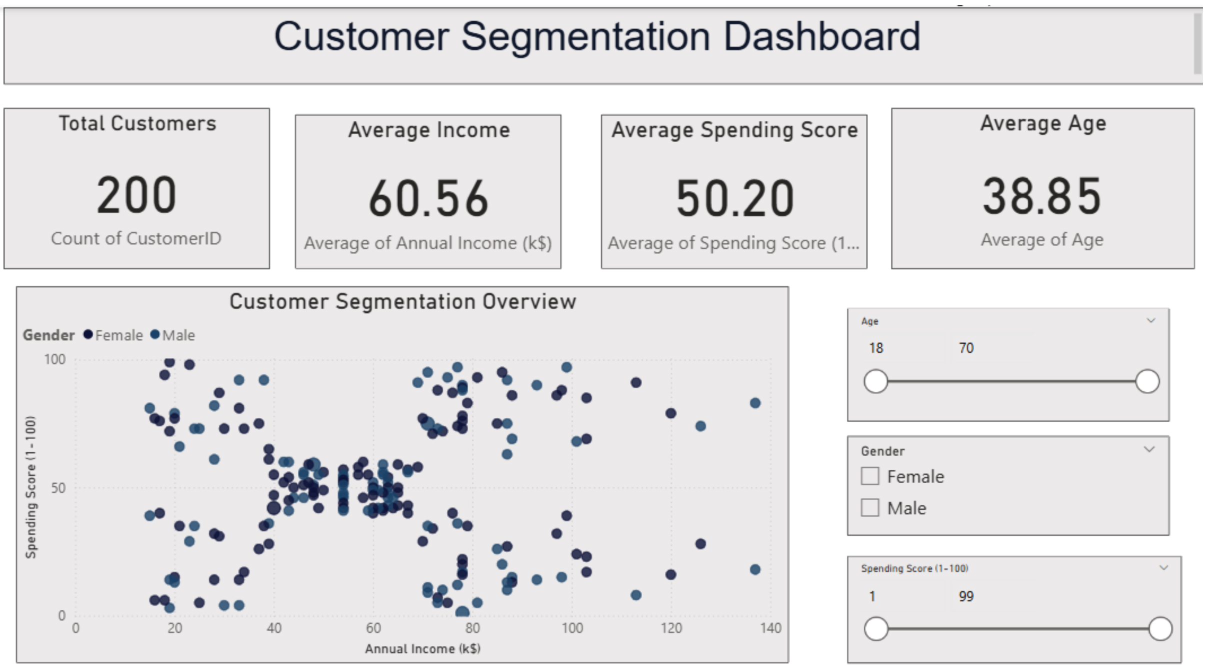
• Slicers for Age and Income filtering

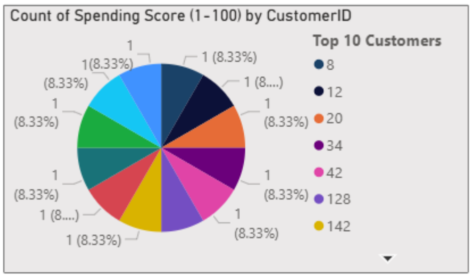
Key Observations:

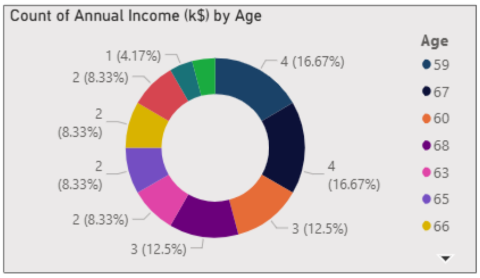
• Customers form clear groupings based on income and spending behavior.

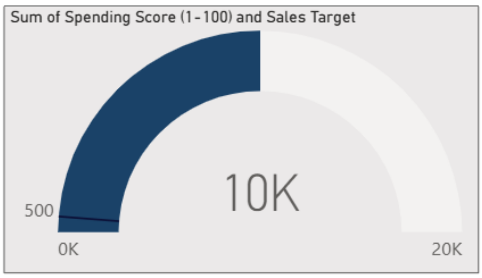
• A major cluster of customers earning between $40k–$70k shows moderate and stable spending patterns.

• Spending behavior is more strongly influenced by spending score than income in certain clusters.









4. BEHAVIORAL SEGMENTATION (K-MEANS CLUSTERING)

K-Means clustering was used to segment customers based on Age and Spending Score, identifying 5 distinct customer groups:

Cluster 1: High-Value Youth (Premium Segment)

- Age: 18–40
- Spending Score: 80–100
- Highly active and premium customers

Cluster 2: Growth-Oriented Youth

- Age: 18–40
- Spending Score: 60–80
- Potential high-value customers

Cluster 3: Budget-Conscious Youth

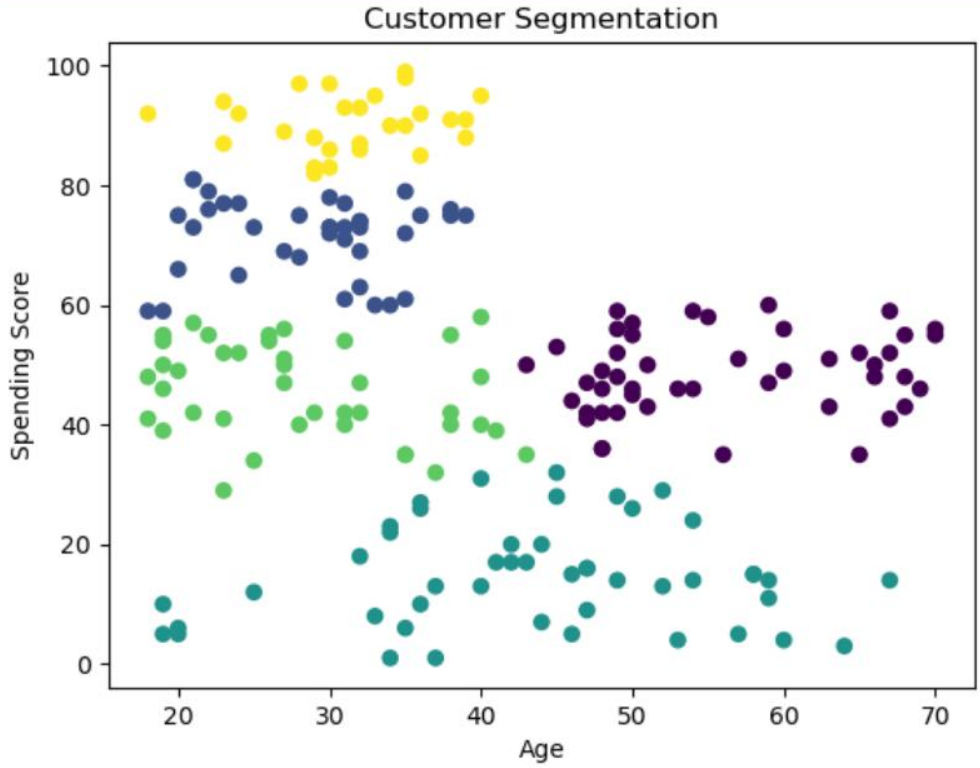
- Age: 18–40
- Spending Score: 30–60
- Moderate engagement level

Cluster 4: Loyal Mature Customers

- Age: 40–70
- Spending Score: 35–60
- Stable and consistent buyers

Cluster 5: At-Risk / Low Engagement Customers

- Age: 18–70
- Spending Score: 0–30
- Low activity and minimal engagement



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5. STRATEGIC BUSINESS RECOMMENDATIONS

High-Value Youth (Premium Segment):

Focus on exclusive offers, premium product launches, influencer campaigns, and personalized digital marketing to maximize revenue.

Growth-Oriented Youth:

Encourage conversion into premium customers through loyalty programs, discounts, and personalized recommendations.

Budget-Conscious Youth:

Use price-sensitive strategies such as seasonal offers, bundle deals, and targeted promotions.

Loyal Mature Customers:

Maintain engagement through loyalty rewards, personalized communication, and consistent service quality.

At-Risk Customers:

Re-engage through aggressive promotions, feedback campaigns, and reactivation discounts.

6. CONCLUSION

This project successfully demonstrates the use of Power BI and Python-based K-Means clustering for customer segmentation.

By transforming raw customer data into actionable insights, the business can shift from generalized marketing to targeted, data-driven strategies. This improves customer engagement, retention, and overall business performance.